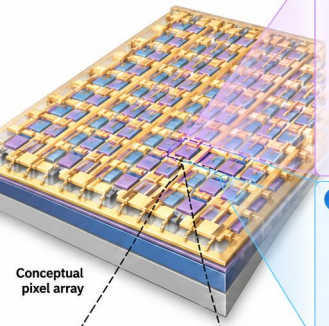
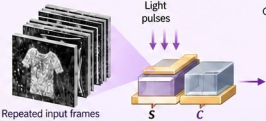
One pixel, two states, weighted computation

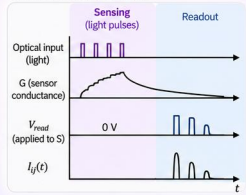


1 Zero-bias temporal sensing

Light history encoded as volatile photoconductance G



Schematic time responses



2 Capacitor weight storage

Store capacitor voltage V_C with fixed V_{in} and variable writing time t_{write} (does not reprogram G)



TG1 ON (weight write)

TG2 OFF

3 Weighted readout and summation

Apply stored V_C to sensor, obtain $I_{ij}(t_r) \approx G_{ij}(t_r)\, V_{C,ij}$ (ohmic readout, small droop)

TG1 OFF

TG2 ON (readout)

